

Open-Ended Lab Problem

Supermarket Management System using Data Structures

Problem Statement

A supermarket wants to develop a small management system to organize products, customers, and billing operations efficiently. Design and implement a Java application that applies different Data Structures and Algorithms concepts learned in the course.

Tasks

Task 1: Product Management using Classes and Arrays

Task:

Create Product and Customer classes and store product records using arrays.

Description:

Apply OOP concepts such as classes, objects, constructors, and arrays to manage supermarket records.

Task 2: Customer Billing Queue

Task:

Implement a queue to manage customers waiting at the billing counter.

Description:

Use Queue operations to serve customers in First In First Out (FIFO) order.

Task 3: Undo Billing using Stack

Task:

Implement a stack to undo the most recent billing transaction.

Description:

Apply Stack operations to demonstrate Last In First Out (LIFO) behavior.

Task 4: Product Searching and Sorting

Task:

Implement searching and sorting operations on product records.

Description:

Compare different searching and sorting techniques for managing supermarket products efficiently.

Task 5: Inventory Management using Linked List

Task:

Implement a linked list to dynamically manage product inventory.

Description:

Perform insertion, deletion, traversal, and search operations using linked lists.

Task 6: Product Management using BST

Task:

Implement a Binary Search Tree to insert, search, and delete product records.

Description:

Use BST operations for efficient hierarchical storage and retrieval of product data.

Additional Requirements

- Use menu-driven implementation.
- Handle invalid inputs properly.
- Display proper output after every operation.
- Add at least one additional feature of your choice.

Expected Outcome

Students will integrate multiple DSA concepts into a single real-world application and understand the practical implementation of different data structures and algorithms.